

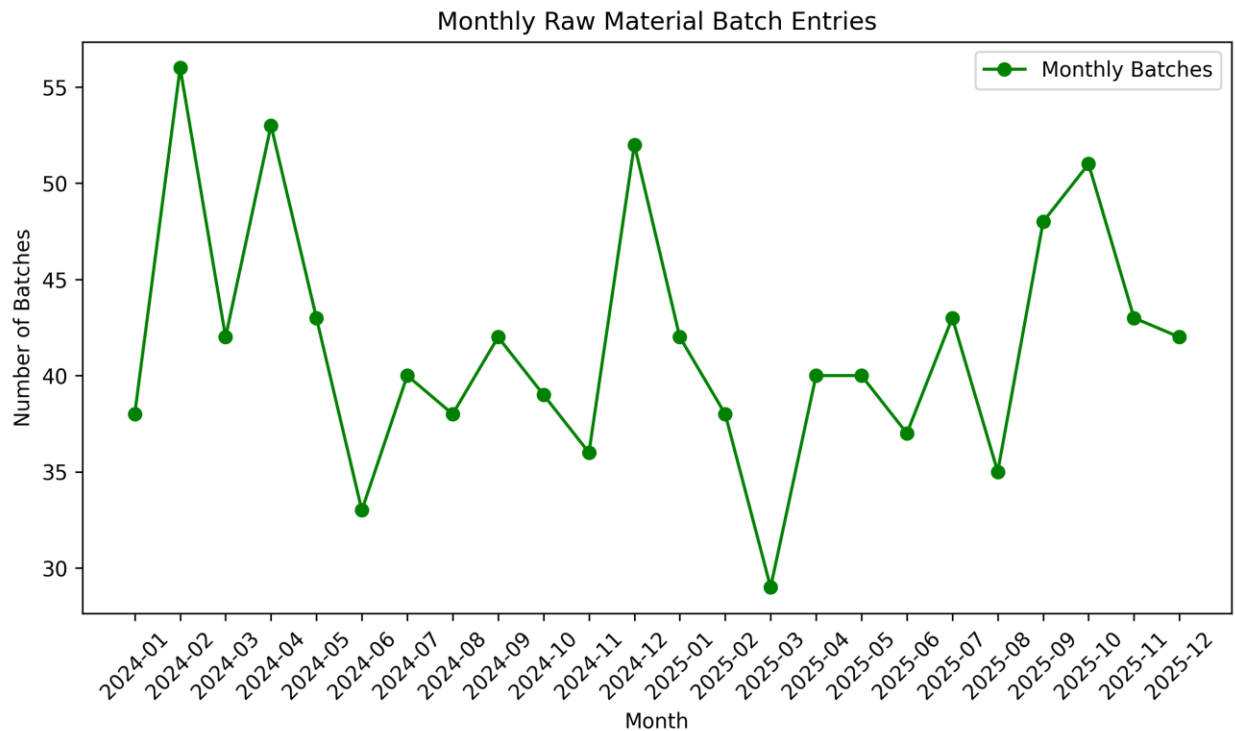


Fig. 1.1 Line Chart

Narrative 1 — Line Chart

Monthly Raw Material Batch Entries

This chart shows the monthly trend in raw material batch entries recorded in the HimShakti inventory dataset. Each point on the line represents the total number of raw material batches entered during a particular month. The purpose of this chart is to understand how inventory intake changes over time and whether there are any visible fluctuations in the volume of raw material entries. By observing these monthly movements, the business can identify patterns in procurement and stock inflow.

A key observation from the chart is that the number of batch entries does not remain constant across all months. Some months show noticeably higher inventory intake, while others reflect relatively lower activity. This variation may indicate differences in procurement cycles, production planning, supplier schedules, or demand-related adjustments in stock intake.

From a business perspective, this insight is useful for improving procurement planning and storage management. If certain months consistently show higher inflow, the business can prepare warehouse space, labour allocation, and expiry monitoring efforts in advance to reduce waste and avoid stock mismanagement.

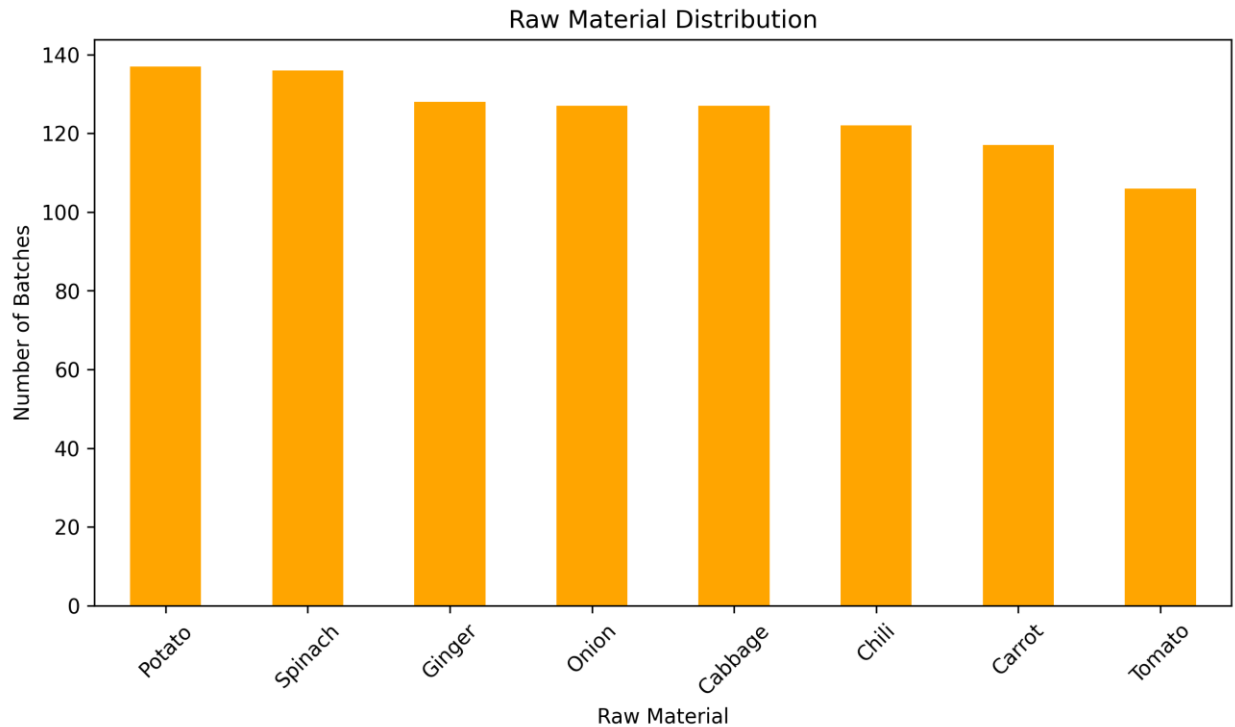


Fig. 1.2 Bar Chart

Narrative 2 — Bar Chart

Monthly Raw Material Batch Entries

This chart presents the distribution of raw materials across the inventory dataset by showing how frequently each raw material appears in recorded batches. The x-axis represents the raw material categories, while the y-axis shows the number of batches associated with each material. The chart is designed to answer a key business question: which raw materials are most commonly handled in the inventory process and therefore require closer stock monitoring and replenishment planning.

One clear observation from the chart is that some raw materials appear much more frequently than others, indicating that they are core inputs in the production process. These high-frequency materials likely have a greater impact on operational continuity because any shortage, overstocking, or expiry issue related to them would affect a larger share of production activity than less frequently used materials.

For the business, this insight is highly valuable because it helps prioritize inventory control efforts. Raw materials with the highest usage frequency should be tracked more carefully for reorder levels, storage conditions, and expiry dates. This can support better purchasing decisions, reduce avoidable waste, and improve production stability.

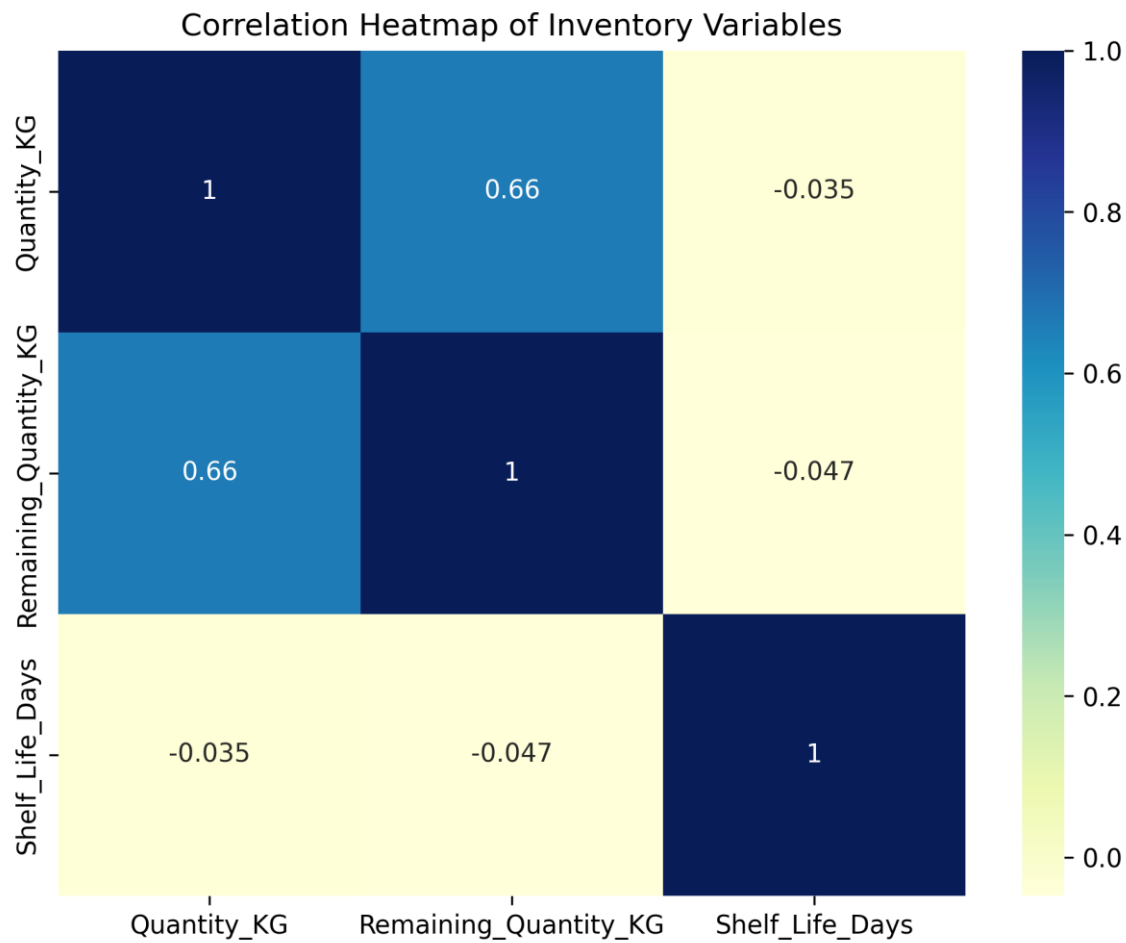


Fig. 1.3 Heap Chart

Narrative 3 — Heatmap

Correlation Heatmap of Inventory Variables

This heatmap shows the correlation between key numerical variables in the HimShakti inventory dataset, such as quantity, remaining quantity, and shelf life. Correlation values help measure the strength and direction of relationships between variables. A value closer to 1 indicates a strong positive relationship, a value closer to -1 indicates a strong negative relationship, and values near 0 suggest little or no direct relationship. This visualization helps summarize multiple numerical relationships in a compact and interpretable format.

An important observation from the heatmap is that quantity and remaining quantity are likely to show a positive relationship, which is expected because larger incoming batches may naturally leave behind larger remaining stock if consumption is not equally fast. If shelf life shows weak or limited correlation with quantity-related fields, it suggests that expiry risk depends not only on stock size but also on timing and stock movement.

For the business, these findings support better inventory control by highlighting which variables should be monitored together. Understanding these relationships can improve expiry-risk analysis, stock planning, and waste-reduction decisions.

